

1. **Configure your hostname, ip address, gateway and dns.**
 - a. Hostname: idtech
 - b. IP address: 192.168.x.150/24 (x sizin şəbəkənin NAT etdiyi IP adresidir, hər kəsdə fərqli ola bilər)
 - c. Gateway: Look your current gateway
 - d. DNS: 8.8.8.8
2. **You must connect your OS only via SSH and attached IP address.**
3. **Configure YUM repos.** Do it by mounting your sr0 device to mnt folder. (2 repos: 1st is Base and 2nd is AppStream). Install bash-completion.
4. **Set default values for new users.** Set the default password validity to 90 days, and set the first UID that is used for new users to 2000.
5. **Create User accounts with supplementary (secondary) group.**
 - Create the group a named "sysadms".
 - Create users as named "natasha" and "harry", will be the supplementary group "sysadms".
 - Create a user as named "sarah", should have non-interactive shell and it should be not the member of "sysadms".
 - Password for all users should be "trootent"
6. **Configure a cron job:**
 - a) that runs every minutes and executes: logger "EX200 in progress" as the user natasha;
 - b) that runs every Sunday, at 16:00 every 30 sec: "echo `date` >> /root/file.txt" as the user root.
7. **Create the Directory "/home/manager" with the following characteristics.**
 - Group ownership of "/home/manager" should go to "sysadms" group.
 - The directory should have full permission for all members of "sysadms" group but not to the other users except "root".
8. **Copy the file /etc/fstab to /var/tmp/ and configure the "ACL" as mentioned following.**
 - The file /var/tmp/fstab should be owned by the "root".

- The file /var/tmp/fstab should belong to the group "root".
- The file /var/tmp/fstab should not be executable by any one.
- The user "sarah" should be able to read and write to the file.
- The user "harry" can neither read nor write to the file.
- Other users (future and current) should be able to read /var/tmp/fstab.

9. Find files owned by harry and copy it to /opt/dir folder.

10. Create user 'bob' with 2112 uid and set the password 'trootent'.

11. Create an archive '/root/backup.tar.bz2' of /usr/local directory and compress it with bzip2.

12. Find the rows that contain only "Port" or "port" from /etc/ssh/ and write result to the /tmp/newfile.

13. Customize tuned profile and set powersave.

14. Create a logical Volume and mount it permanently:

- Create 1Gb volume group named idtechgroup with 8 MB PE size. Create the logical volume with the name "idtech" by using 50PE's from the volume group "idtechgroup".

- Mount it on /tmp/idtech with file system xfs.

15. Resize a logical Volume:

Resize the logical volume "idtech" so that after reboot the size should be in between 400MB to 500MB.

16. Add a swap partition of 512MB and mount it permanently.

17. Create LVM-VDO named "vdo1" with logical size 50Gb and mount it.

22. Configure Recurring jobs

Note: For This Question You Need To Install Lookup Package (mod_lookup_identity).

- a) Create a script `/usr/local/bin/rhcsa_capture` that lists all files in `/home` and saves the output to `/root/log_output/home_logs.trc`. Make sure the script is executable
- b) Create a systemd service unit **`rhcsa_capture.service`** that runs the script
- c) Create a systemd timer unit **`rhcsa_capture.timer`** that triggers the service every 1 minutes.
- d) Enable and start the timer so it begins working immediately and continues after reboots.
- e) Verify the timer is active and confirm that `/root/log_output/home_logs.trc` is updated